

APUKA, Russian Federation

WMO: 259560

Lat: 60.443N

Lon: 169.608E

Elev: 5

StdP: 101.27

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.6	-24.6	-32.2	0.2	-26.0	-30.2	0.2	-23.8	20.5	-7.6	19.1	-8.8	4.5	50	0.854

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.2	17.3	13.8	15.9	12.9	14.8	12.4	14.3	16.4	13.4	15.3	12.6	14.3	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	9.4	15.5	12.2	8.9	14.5	11.6	8.5	13.7	40.2	16.5	37.7	15.5	35.7	14.4	17.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.5	14.2	12.3	DB	-30.3	20.7	3.8	1.8	-33.0	21.9	-35.2	22.9	-37.4	23.9	-40.1	25.2	
			WB	-30.5	16.4	3.7	1.0	-33.2	17.1	-35.4	17.6	-37.5	18.2	-40.2	18.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-13.0	-12.6	-10.2	-4.6	2.2	7.5	11.7	11.9	8.1	0.6	-5.9	-11.1
	DBStd	10.27	7.32	6.45	6.49	4.29	2.16	2.07	2.24	1.74	2.52	4.10	5.77	6.41
	HDD10.0	4223	712	632	627	437	242	77	9	3	64	291	477	653
	HDD18.3	7129	970	865	886	687	501	323	207	198	306	549	727	911
	CDD10.0	136	0	0	0	0	0	4	61	63	8	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.0	7.1	6.8	6.3	4.7	3.3	3.2	3.2	3.4	3.9	5.0	6.3	7.2
Precipitation	PrecAvg	465	29	23	34	20	19	35	33	65	60	62	52	34
	PrecMax	688	91	65	71	45	71	135	125	157	136	141	109	79
	PrecMin	300	0	0	1	7	5	7	3	0	10	0	9	1
	PrecStd	93	25	16	22	10	14	28	26	35	33	37	31	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.4	2.4	2.1	5.0	10.6	16.1	20.0	18.6	15.6	10.1	4.7	2.6
		MCWB	1.2	0.9	0.9	3.2	7.6	11.3	15.0	14.7	12.4	7.3	3.8	1.5
	2%	DB	1.3	0.3	0.8	3.3	8.3	13.2	18.1	16.8	13.8	8.5	3.6	1.5
		MCWB	0.1	-0.9	-0.3	1.5	6.2	10.7	14.4	13.4	10.9	6.3	2.9	0.4
	5%	DB	-0.1	-1.1	-0.1	2.0	6.8	11.8	16.4	15.6	12.7	7.2	2.6	0.4
		MCWB	-1.2	-2.0	-0.9	0.4	5.1	9.6	13.4	12.8	10.4	5.5	1.6	-0.6
	10%	DB	-2.5	-2.8	-1.0	1.2	5.5	10.6	15.2	14.7	11.8	6.1	1.6	-1.4
		MCWB	-3.6	-3.9	-1.9	-0.2	3.8	8.7	12.8	12.3	9.8	4.3	0.5	-2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.1	1.1	3.4	7.9	12.5	16.2	15.2	13.1	7.9	4.0	1.7
		MCDB	2.1	2.4	1.8	4.8	10.3	14.8	19.0	18.1	15.1	9.6	4.4	2.3
	2%	WB	0.3	-0.7	0.1	1.6	6.5	10.8	14.9	13.9	11.6	6.6	2.9	0.5
		MCDB	1.2	0.1	0.5	2.9	8.2	12.8	17.2	16.0	13.2	8.2	3.4	1.3
	5%	WB	-1.3	-2.1	-0.8	0.6	5.2	9.8	13.9	13.2	10.8	5.7	1.6	-0.6
		MCDB	-0.2	-1.3	-0.1	1.8	6.6	11.6	15.8	15.1	12.2	7.2	2.3	0.4
	10%	WB	-3.6	-3.7	-1.9	-0.1	4.0	9.0	13.0	12.5	10.1	4.6	0.6	-2.4
		MCDB	-2.7	-2.8	-1.0	1.1	5.3	10.4	14.8	14.3	11.5	6.0	1.5	-1.4
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.8	6.8	6.1	4.2	3.9	4.2	4.2	5.0	5.0	4.7	5.1
		MCDBR	4.3	6.7	4.8	5.2	6.0	6.2	6.7	6.1	6.3	5.0	4.3	4.5
		MCWBR	4.2	6.5	4.7	4.5	4.7	4.7	4.5	4.6	5.0	4.2	4.2	4.8
	5% WB	MCDBR	4.4	6.1	4.5	4.5	6.0	5.7	5.8	5.3	5.3	4.7	4.2	4.8
		MCWBR	4.5	6.2	4.6	4.1	4.8	4.5	4.5	4.2	4.5	4.2	4.2	5.2
Clear-Sky Solar Irradiance	taub	0.236	0.284	0.336	0.383	0.434	0.406	0.435	0.428	0.340	0.314	0.266	0.218	
	taud	2.077	2.130	2.040	1.963	1.986	2.252	2.206	2.194	2.456	2.314	2.106	1.988	
	Ebn at Noon	520	696	765	795	783	822	783	752	784	663	459	366	
	Edh at Noon	42	74	113	148	157	123	126	118	77	62	41	29	
All-Sky Solar Radiation	RadAvg	0.37	1.12	2.64	4.49	5.00	4.95	4.97	3.65	2.52	1.28	0.41	0.16	
	RadStd	0.06	0.15	0.40	0.37	0.60	0.58	0.44	0.42	0.30	0.14	0.08	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page